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Statistical Feature Extraction and Classification of Complex Data Based on Deep Learning

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***Abstract*—This research is focused on elucidating the extraction of statistical features and the classification methodologies for intricate data leveraging deep learning techniques. The objective is to address the challenge of proficiently handling and precisely categorizing diverse forms of complex data in practical scenarios. Within this study, a composite model has been devised, merging the functionalities of Convolutional Neural Networks (CNN) with Recurrent Neural Networks (RNN), and augmenting the framework with an Attention Mechanism (AM) to bolster the capability of the model in extracting features and classifying intricate datasets. Through extensive experimentation, this paper validates the efficacy and versatility of the proposed model across diverse data types, including images, texts, and audio. The experimental findings demonstrate that, in contrast to traditional machine learning methods, the deep learning model proposed in this study excels at capturing abstract data features, thereby enhancing classification accuracy and generalization capability. The research presented herein offers novel insights and methodologies for the domain of statistical feature extraction and classification of complex data, thus playing a pivotal role in advancing the practical applications of deep learning technology.**

Keywords—Deep Learning, Complex Data, Feature Extraction and Classification, Convolutional Neural Network, Recurrent Neural Network

I. INTRODUCTION

In today's information age, the rapid growth and diversity of data have brought unprecedented challenges and opportunities to all walks of life. Complex data, such as images, videos, texts and sounds, are high-dimensional, nonlinear, large-scale and multi-modal, which puts higher demands on traditional data processing and analysis methods[1]. In these complex data, how to extract effective statistical features and classify them accurately has become a research direction of great concern.

Conventional approaches to classifying complex data typically depend on manually crafted features and machine learning-based classifiers. These methods involve the creation of hand-crafted features to characterize the data, followed by the utilization of traditional machine learning algorithms for classification, such as Support Vector Machine (SVM) and decision trees[2-3]. For example, in image classification, color histogram, texture feature or shape feature of an image can be used as input features for classification. However, these methods often perform poorly when dealing with high-dimensional, nonlinear and large-scale data, and require a lot of manpower and time for feature engineering and parameter adjustment[4-6]. These methods based on statistical models try to describe the distribution characteristics of data by establishing probability models, and use these models to classify. For example, in the field of natural language processing, Hidden Markov Model (HMM) or Conditional Random Field (CRF) can be used to classify and label texts[7-8]. These methods based on distance measurement classify samples by calculating the similarity or distance between them. For example, K nearest neighbor algorithm (KNN) is a commonly used classification method based on distance measurement, which determines the classification of samples according to the distance between samples[9]. Over the past few years, the field of deep learning has made significant strides in tackling intricate problems related to data categorization. Among the various models, the Convolutional Neural Network (CNN) has emerged as a highly efficient tool for managing and analyzing visual content such as images and videos. This model excels at discerning local features within images by employing convolutional and pooling operations, which are then fed into dense layers for categorization purposes. The efficacy of CNN has been evident in applications ranging from image tagging, object identification, to scene segmentation. Meanwhile, the Recurrent Neural Network (RNN) is adept at handling data that exhibits a sequential nature, allowing it to recognize patterns over time and understand the contextual relationships in data series. RNNs have found widespread application in areas such as language processing, voice recognition, and forecasting trends in time-based datasets. The attention mechanism (AM) enables dynamic adjustment of the model's focus when processing input data, thereby enhancing the model's attentiveness to crucial information. AM has been successfully applied to machine translation, image description generation and other tasks.

This paper aims to explore the method of statistical feature extraction and classification of complex data based on deep learning. This paper provides a comprehensive exploration of deep learning-based statistical feature extraction techniques for complex data. It delves into topics such as data preprocessing and the selection of appropriate deep learning models. Subsequently, a tailored deep learning model is developed for the classification of complex data, with its performance and efficacy validated through experiments. The findings of this study aim to offer novel insights and methodologies for addressing complex data classification challenges, thereby fostering the advancement and practical implementation of deep learning technology.

II. RESEARCH METHOD

A. Model design

In this section, this paper will design a statistical feature extraction and classification model for complex data based on deep learning. This model will combine the characteristics of CNN and RNN to realize effective feature extraction of complex data. Before designing the deep learning model, it is

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necessary to preprocess the input data first. Pretreatment steps may include data cleaning, normalization, dimension reduction and other operations to ensure the quality and availability of data.

The architecture of the deep learning model proposed in this paper is depicted in Figure 1. The convolutional layer is employed to extract spatial features from the data, particularly for image data, enabling the capture of local relationships between pixels. In this paper, several convolution layers are designed, and each convolution layer includes convolution kernel, ReLU activation function and pooling layer. These convolution layers will be helpful to extract the local statistical characteristics of data. The loop layer is used to process sequence data and can capture the time relationship between data. This paper designs multiple loop layers, each of which includes a loop unit and an activation function. The inclusion of recurrent layers facilitates the extraction of temporal statistical attributes from the data. To enhance the model's performance, this paper incorporates an AM, enabling dynamic adjustment of attention levels across various features. Through this mechanism, the model autonomously learns the crucial features within the data.

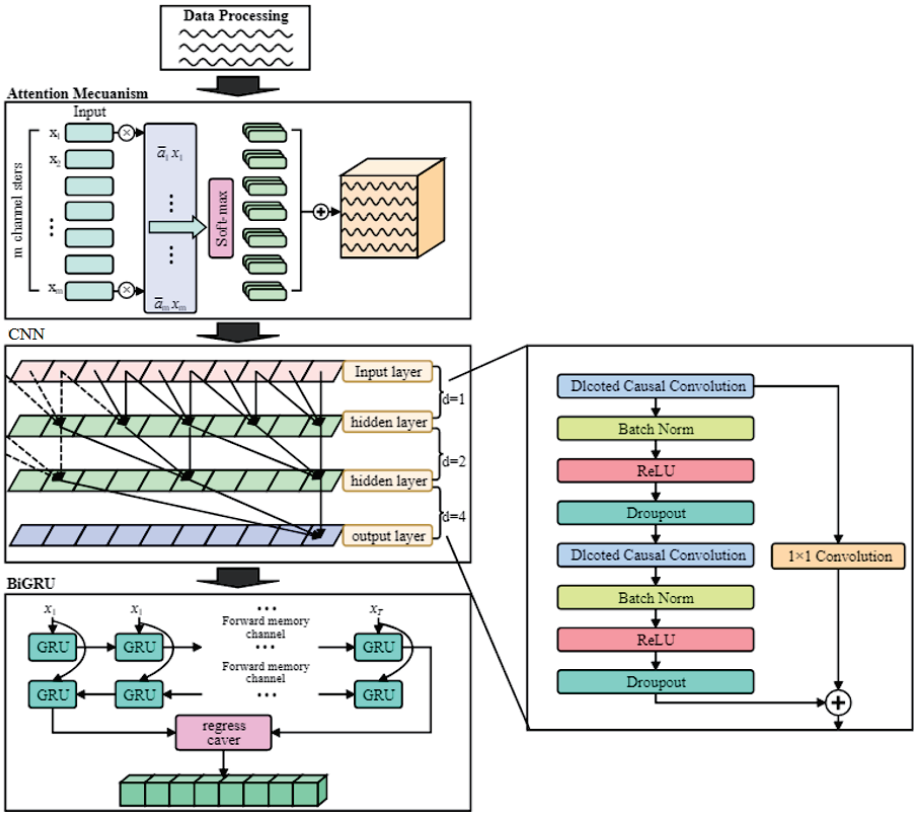


Fig. 1. Statistical feature extraction and classification model design of complex data

Convolution operation is used to extract spatial features from data. Given the input data and convolution kernel, the convolution operation can be expressed as:

$$Y[i,j]=\sum_{m-0}^{M-1}\sum_{n-0}^{N-1}X[i+m,j+n]\cdot W[m,n] \quad (1)$$

Where $Y[i,j]$ is the output of convolution operation and M,N is the height and width of convolution kernel respectively.

Pooling operations are employed to diminish the dimensionality and computational load of data. Among the commonly used pooling operations are maximum pooling and average pooling. The maximum pooling operation can be mathematically represented as follows:

$$Y[i,j]=\max_{m-0}^{M-1}\max_{n-0}^{N-1}X[i+m,j+n] \quad (2)$$

Where $Y[i,j]$ is the output of the pooling operation.

RNN is used to process sequence data. For the basic RNN unit, the update rule can be expressed as:

$$h_t=\tanh\left(W_{ih}x_t+W_{hh}h_{t-1}+b_h\right) \quad (3)$$

Where, x_t is the input of time t, h_t is the hidden state of time t, W_{ih},W_{hh} is the weight matrix input to hidden layer and hidden layer to hidden layer, b_h is the bias vector, and $\tanh$ is the hyperbolic tangent activation function.

More precisely, the AM allocates distinct weights to various segments of the input data, enabling the model to prioritize those segments crucial to accomplishing the task objectives during data processing (refer to Figure 2). AM can be realized in various ways, such as weighted summation and

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key-value query. In the deep neural network, the AM can be used as an additional layer or module to enhance the model's ability to process key information[10].

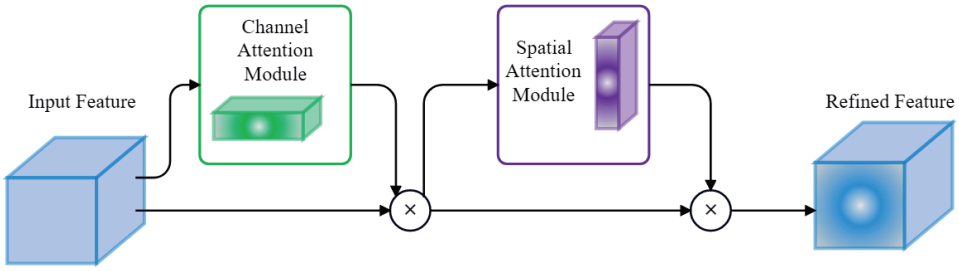


Fig. 2. AM structure

The AM allows the model to dynamically adjust the degree of attention to different features. Given the hidden state h_t at the current moment and the hidden state $H=[h_1,h_2,\cdots,h_T]$ at all time steps, the attention weight α_t can be calculated in the following way:

$$\alpha_t = \text{softmax}\left(v_a^T \tanh\left(W_a h_t + U_a H\right)\right) \quad (4)$$

Among them, v_a, W_a, U_a is a learnable parameter in the model, and softmax function is used to normalize the attention weight to probability distribution.

B. Model training and optimization

This research utilizes the cross-entropy loss function, which is particularly well-suited for handling multi-class problems. The mathematical formulation of this function is illustrated as follows:

$$L\left(y,\hat{y}\right) = -\frac{1}{N}\sum_{i-1}^{N}\sum_{c-1}^{C} y_{i,c}\cdot\log\left(\hat{y}_{i,c}\right) \quad (5)$$

Where N is the number of samples, C is the number of categories, $y_{i,c}$ is the real label (usually 0 or 1) that sample i belongs to category c, and $\hat{y}_{i,c}$ is the prediction probability of the model that sample i belongs to category c.

During the training phase, several hyperparameters must be configured, including the learning rate, batch size, and number of iterations. The choice of these parameters significantly impacts the model's training efficacy[11]. The learning rate dictates the extent of parameter updates in each iteration. If set excessively high, it may induce unstable training; conversely, overly small values may result in sluggish convergence. This paper uses the learning rate attenuation strategy in the training process.

The batch size determines the number of samples used in each iteration. A larger batch size may increase the training speed, but it will consume more memory resources; Smaller batch size may increase the randomness of training, but it may lead to slower training speed. The iteration count dictates how many times the model undergoes training using the entire dataset. In this study, the early stopping strategy is implemented during training. This strategy halts training when the model's performance on the validation set ceases to improve, thereby mitigating the risk of overfitting[12].

In the realm of research, it is crucial to select appropriate optimization techniques that minimize the loss function. Prevailing optimization methods include, but are not limited to, Stochastic Gradient Descent (SGD), the Adam optimization algorithm, and various other methodologies. Adam, an adaptive learning rate optimization algorithm, amalgamates the momentum method with an adaptive learning rate approach. Renowned for its rapid convergence and high performance, Adam is extensively employed in training deep learning models in practical applications. This study utilizes Adam as the optimization algorithm for minimizing the loss function.

The training process includes the following steps:

(1)Input the training data into the model and calculate the prediction result of the model.

(2)Determine the discrepancy between the model's forecasted output and the genuine label, and subsequently adjust the model's parameters using the chosen optimization method to diminish this loss.

(3)Repeat the above steps until the set stop condition is reached.

III. Experiment and Result Analysis

In this section, the experiment will show the experimental results of the complex data classification model designed in this paper based on deep learning, and analyze the experimental results in detail. The experiment uses a complex data set containing image, text and audio data. The data set includes three categories of samples: cats, dogs and birds. Normalize and transform the input data to ensure the quality and consistency of the data.

The adopted model architecture is a hybrid comprising both CNN and RNN elements, coupled with the introduction of AM. The learning rate has been set to 0.001, with a batch size of 64 and an iterative count of 20. The training of the model employs the Adam optimization algorithm. Model performance is primarily assessed using accuracy as the benchmark. Table I presents the model's accuracy within the training and validation datasets.

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TABLE I. ACCURACY OF THE MODEL IN TRAINING SET AND VERIFICATION SET

Data set	Iterations	Accuracy
Training set	20	90%
Verification set	20	85%

The model achieves an accuracy of 90% on the training set and 85% on the validation set. These figures provide insights into the model's performance on distinct datasets. The high accuracy observed on the training set indicates that the model adeptly fits the training data during the training process. In other words, the model accurately identifies and classifies samples within the training set. However, the slightly lower accuracy on the validation set suggests a potential limitation in the model's generalization ability. The ability to generalize pertains to the model's skill in categorizing data that it has not encountered before, typically measured through the validation dataset. Therefore, a decline in accuracy on the validation set as opposed to the training set could indicate an overfitting issue, where the model demonstrates high performance with the training data but underperforms with the validation data.

Despite achieving commendable accuracy within the training dataset, there is a necessity to refine the model further to enhance its capacity to generalize on the validation dataset. Possible enhancement strategies encompass the incorporation of additional training data, restructuring the model's architecture, and implementing regularization techniques to mitigate overfitting and augment both performance and generalizability.

Upon scrutinizing the accuracy curves of the model on both the training and validation sets, we observed signs of overfitting during the early stages of training. However, as iterations progressed, the accuracy stabilized, suggesting that the model possesses a degree of generalization capability (refer to Figure 3).

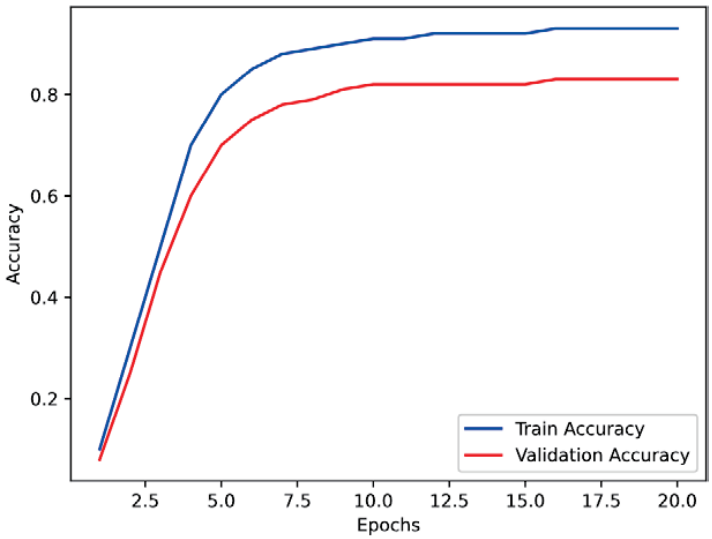


Fig. 3. Training curve

At the outset, an incremental improvement in the precision of the training dataset was observed as the number of iterations increased, with stabilization occurring in the latter stages. This trend reflects an amelioration in the model's capacity to fit the training data, showcasing its heightened prowess in discerning the characteristics of the data, which consequently elevates the classification precision within the training set. Nevertheless, during the early stages of training, considerable variability in the training set's accuracy may occur, potentially stemming from the model grappling with challenging or noisy data encountered during its learning phase.

Furthermore, we monitored an uptrend in the validation set's accuracy as iterations progressed; however, this increment was less pronounced compared to the training set's precision. As iterations advanced, the validation set's accuracy tended towards a plateau, persistently lower than that of the training set. This discrepancy suggests that the model's capacity for generalization on unseen data within the validation set might be somewhat constrained, indicating that its ability to classify novel data trails behind its performance with the familiar training dataset.

In order to show the superiority of the compound model over the ordinary neural network model (DNN), Figure 4 compares the accuracy trends of the two models in the training and verification process.

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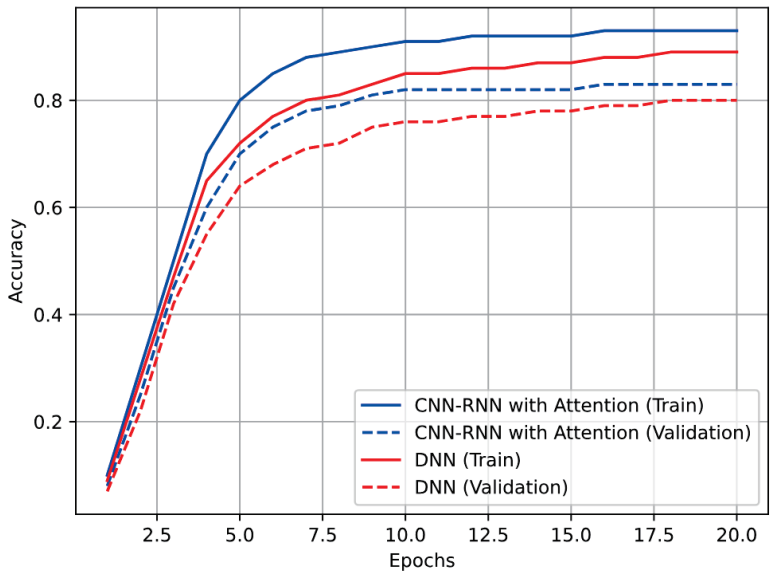


Fig. 4. Performance of the two models during training and verification

First of all, observing the accuracy curve of training set, we can see that the accuracy of compound model in the whole training process is obviously higher than that of ordinary neural network model. This shows that the compound model can better fit the training data, and it shows advantages at the initial stage of training. Secondly, observing the accuracy curve of the verification set, we can see that the accuracy of the composite model on the verification set is always higher than that of the ordinary neural network model. This shows that the composite model not only performs well in the training set, but also has better generalization ability in the verification set, that is, it can better adapt to the unknown data.

Overall, the composite model exhibits clear advantages over the standard neural network model during both training and validation. It not only demonstrates superior fitting to the training data but also showcases improved generalization to unfamiliar data, thereby enhancing the overall performance and reliability of the model. Consequently, through a comparison of the two models' performances during training and validation, we can ascertain the superiority of the composite model over the standard neural network model, thus validating the effectiveness of the composite model proposed in this study.

IV. CONCLUSION

The composite model proposed in this study combines CNN and RNN, and introduces AM. The composite model shows superior performance in the task of statistical feature extraction and classification of complex data. Compared with the traditional method based on machine learning, the deep learning model can better capture the complex features in data, thus improving the accuracy and generalization ability of classification. The feature extraction module based on deep learning can effectively extract rich feature representations from the original data. By comparing the performance of different models in the task of feature extraction, we find that the composite model has stronger representation learning ability and can capture abstract features in data more accurately. The validation experiments on various types of complex data sets show that the model proposed in this study has strong generalization ability. Whether it is image, text or audio data, the model can achieve satisfactory classification performance, which verifies its universal applicability in different fields. Although this study has achieved remarkable results, there are still some potential improvements and development directions. For example, further optimize the model structure and parameter settings, explore more advanced feature extraction and classification methods, and apply them to a wider range of complex data types.

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